LESSON PLAN

EMI/EMC (14)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	37
Total No. of Lesson Plans	46	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Explanation of Duplex servo actuators
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Explanation of Duplex servo actuators.
5. Specific Objectives	14.2.1.1 Trainee will be able to explain the functions of AFCS.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of AFCS block diagram and Duplex Servo schematic; Instructor supervision required for schematic interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the AFCS?
(b) Define a duplex system in the context of flight controls.
(c) Why is redundancy critical in servo actuators?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of Duplex Servo Actuator • Comparison with Single Servo Systems • Basic Hydraulic/Electrical Architecture • Role in AFCS Redundancy	L & Ds	PPT, IB	15 min
(b)	• Operational Logic of Dual Channels • Failure Detection and Isolation • Load Sharing Mechanisms • Impact on Flight Control Stability	L & Ds	PPT, PDF	20 min
(c)	• Maintenance Implications of Duplex Design • Troubleshooting Common Failures • System Testing Procedures • Integration with Yaw Damping Circuits	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how a duplex servo actuator maintains control authority during a single-channel failure.
(b) Describe the sequence of events when the AFCS detects a discrepancy between duplex channels.

12. Home Assignment and Follow Up	Prepare a brief note on how the principles of duplex servo actuators apply to the torque limiting and phase advance circuits discussed in LP 38.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Purpose and principle of operation of yaw damping circuit, explanation of torque limiting and phase advance.	Refer to the master syllabus reference list	LP 38

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min